

Feasibility-Aware Quantum-Classical Benchmarking for Occlusion-Aware UAV Trajectory Optimization

A HUBO benchmark that separates solution quality from constraint satisfaction

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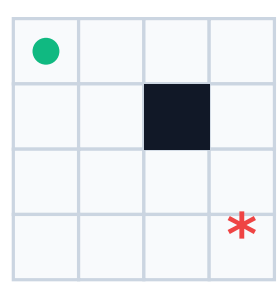
Poster focus. We recast occlusion-aware UAV tracking as a higher-order binary optimization benchmark and evaluate it with feasibility-first quantum-classical metrics.

1. Quantum-compatible model

Occlusion, motion, one-hot, and proximity terms are represented in one HUBO objective.

1 Problem

Goal: plan a UAV trajectory through a cluttered grid while preserving line-of-sight to a target.



- Avoid obstacle cells and invalid moves.
- Keep the target visible whenever possible.
- Reach the terminal cell while satisfying one-hot position constraints.
- Benchmark question: can a quantum-compatible encoding produce feasible, useful trajectories?

Same planning class as continuous multi-convex MPC, reformulated as a discrete quantum-optimization benchmark.

2 HUBO formulation

Binary state: $x[t,v] = 1$ when the UAV occupies grid cell v at time t .

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}$$

Occlusion term
Bresenham line-of-sight checks which obstacles block target visibility.

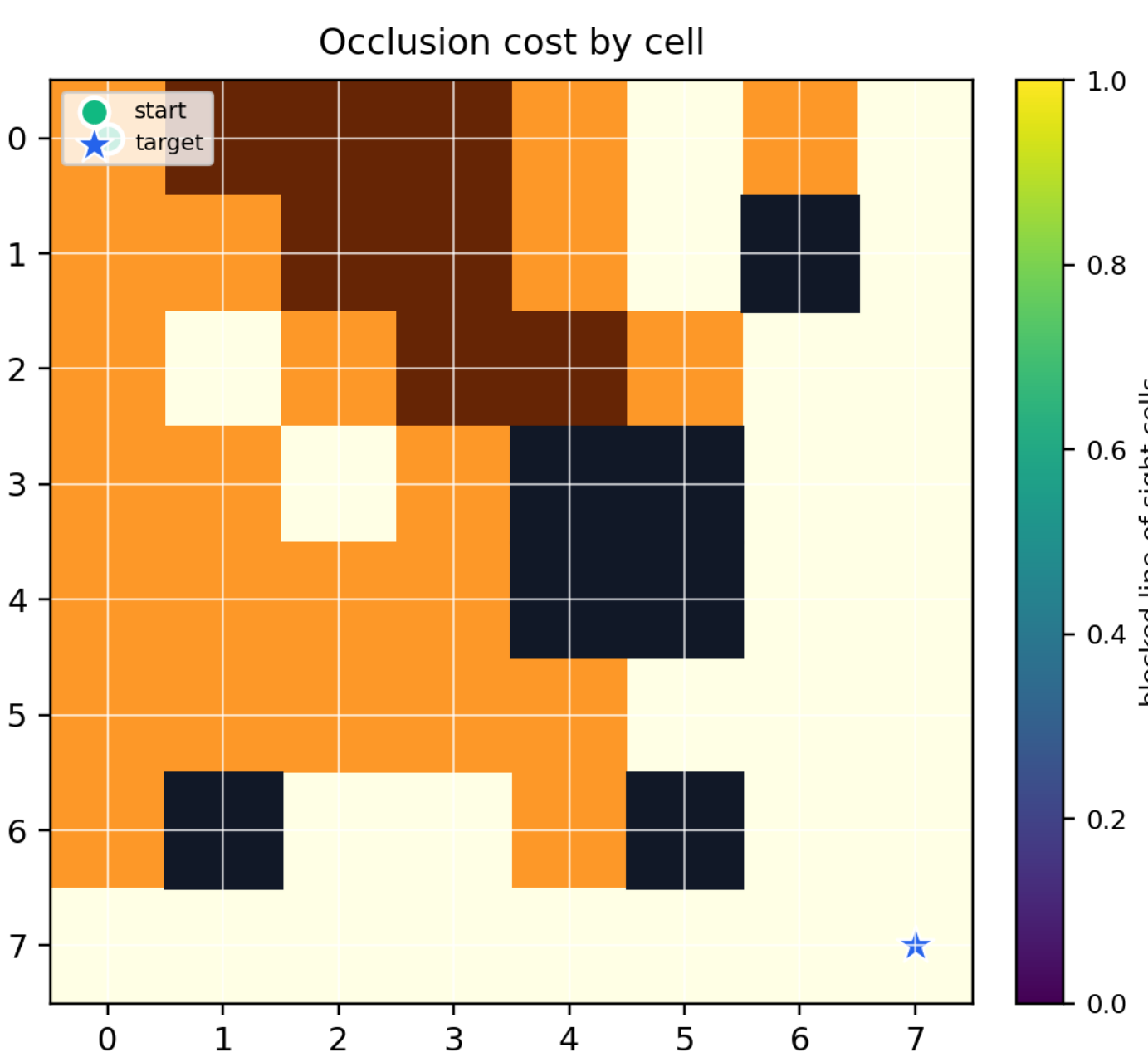
$$H_{\text{occ}} = \lambda_{\text{occ}} \sum x[t,u] |\text{LOS cap } 0|$$

Cubic proximity term
Penalizes lingering in a buffer cell across three adjacent times.

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum x[t-1,v]x[t,v]x[t+1,v]$$

Why HUBO rather than QUBO?
Quadrating the cubic proximity term would add about 3,172 ancillas (about 30% more variables) and create dense couplings that reduce sparsity.

3 Occlusion model



Yellow cells have higher line-of-sight obstruction cost. Black cells are obstacles; the star marks the target.

Design choice: occlusion is modeled at the cell level, so every decoded trajectory can be checked independently after sampling.

4 QCE relevance

- Quantum algorithms: higher-order binary optimization and QAOA-ready formulations.
- Engineering practice: UAV autonomy, sensing, visibility, and planning under constraints.
- Benchmarking: fair comparison between quantum-inspired solvers and classical planners.

2. Feasibility-first metrics

Decoded trajectories are checked directly, rather than trusting raw solver energy alone.

5 Feasibility-aware benchmarking

Raw HUBO energy can be misleading. A solver may lower soft cost while violating hard constraints, so energy and feasibility are reported separately.

- 1 Energy decomposition
Report hard-penalty terms and soft objective terms separately.

- 2 Independent constraint check
Decode every sample and verify one-hot, motion, obstacle, and terminal constraints.

- 3 Matched sampling budget
Use equal reads or runs so solvers face the same search budget.

- 4 Penalty-weight sweep
Expose regimes where low penalties reward infeasible samples.

Key rule: prefer a slightly higher-cost feasible path over a low-energy but invalid sample.

6 Two-tier study design

State-vector QAOA scales as 2^n . A full 8×8 , 960-variable instance is therefore not directly simulatable, so the study uses two tiers.

TIER 1: full 8×8 grid
960 variables
A*, SA, and Neal

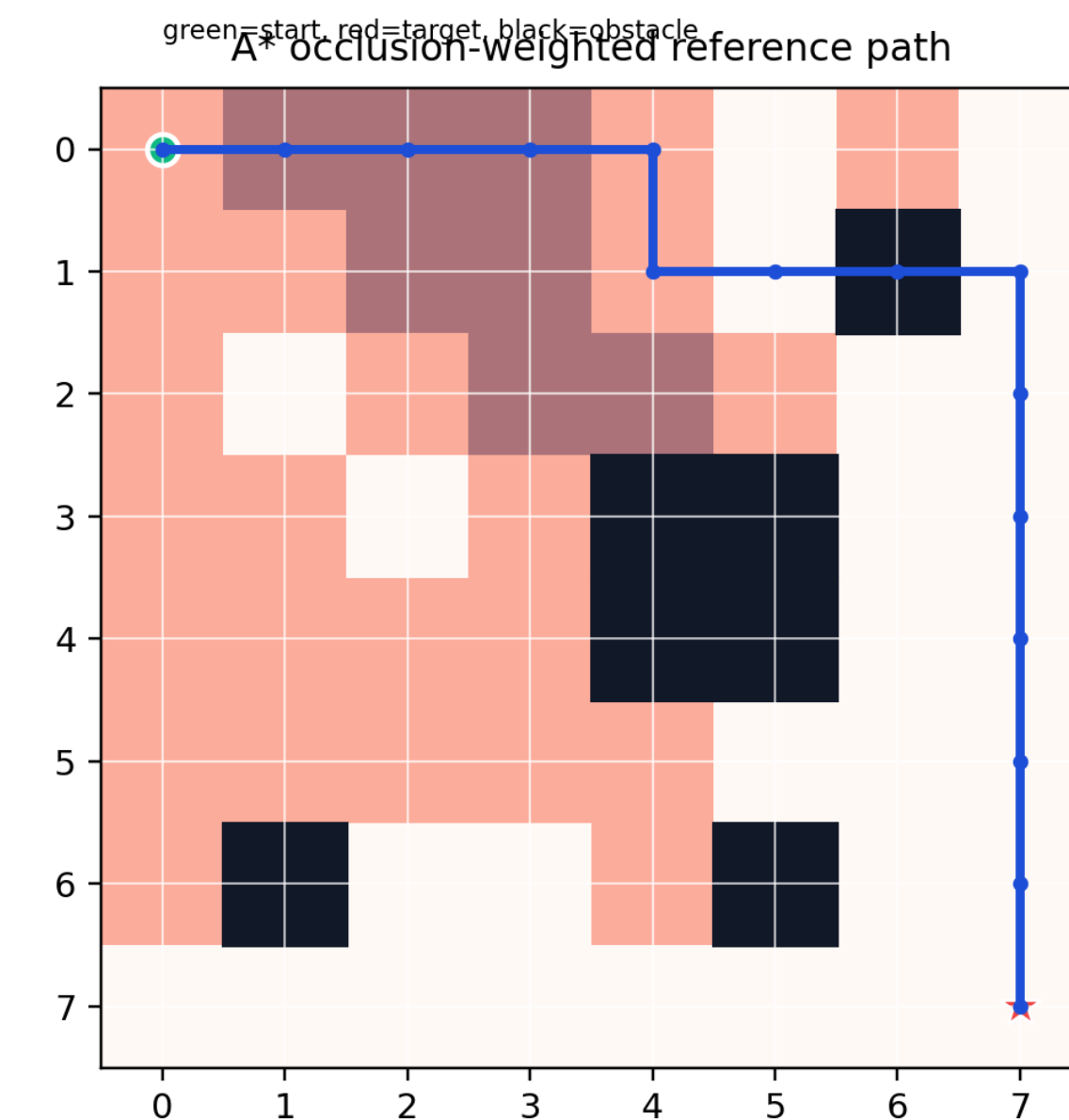
TIER 2: tiny 2×2 grid
8 variables
+ Qiskit QAOA correctness check

Interpretation: Tier 1 tests whether the formulation produces sensible plans. Tier 2 confirms that the quantum encoding reaches the same optimum when exact simulation is possible.

7 Classical A* baseline

A* runs on the same grid using an occlusion-weighted cell cost. This grounds the HUBO benchmark against a conventional trajectory planner.

$$\text{cell cost} = 1.0 + \lambda_{\text{occ}} |\text{LOS cap } 0| + \lambda_{\text{prox}} 1[v \text{ in buffer}]$$



A* reference path: feasible, reaches the target, and maintains about 80% visibility.

8 Main contributions

- A discrete HUBO benchmark for occlusion-aware UAV trajectory optimization.
- A feasibility-first evaluation protocol for quantum and quantum-inspired solvers.
- Evidence that penalty choices strongly affect reported feasibility.

3. Baseline-grounded study

A*, simulated annealing, Neal, and tiny QAOA separate practical value from hardware limits.

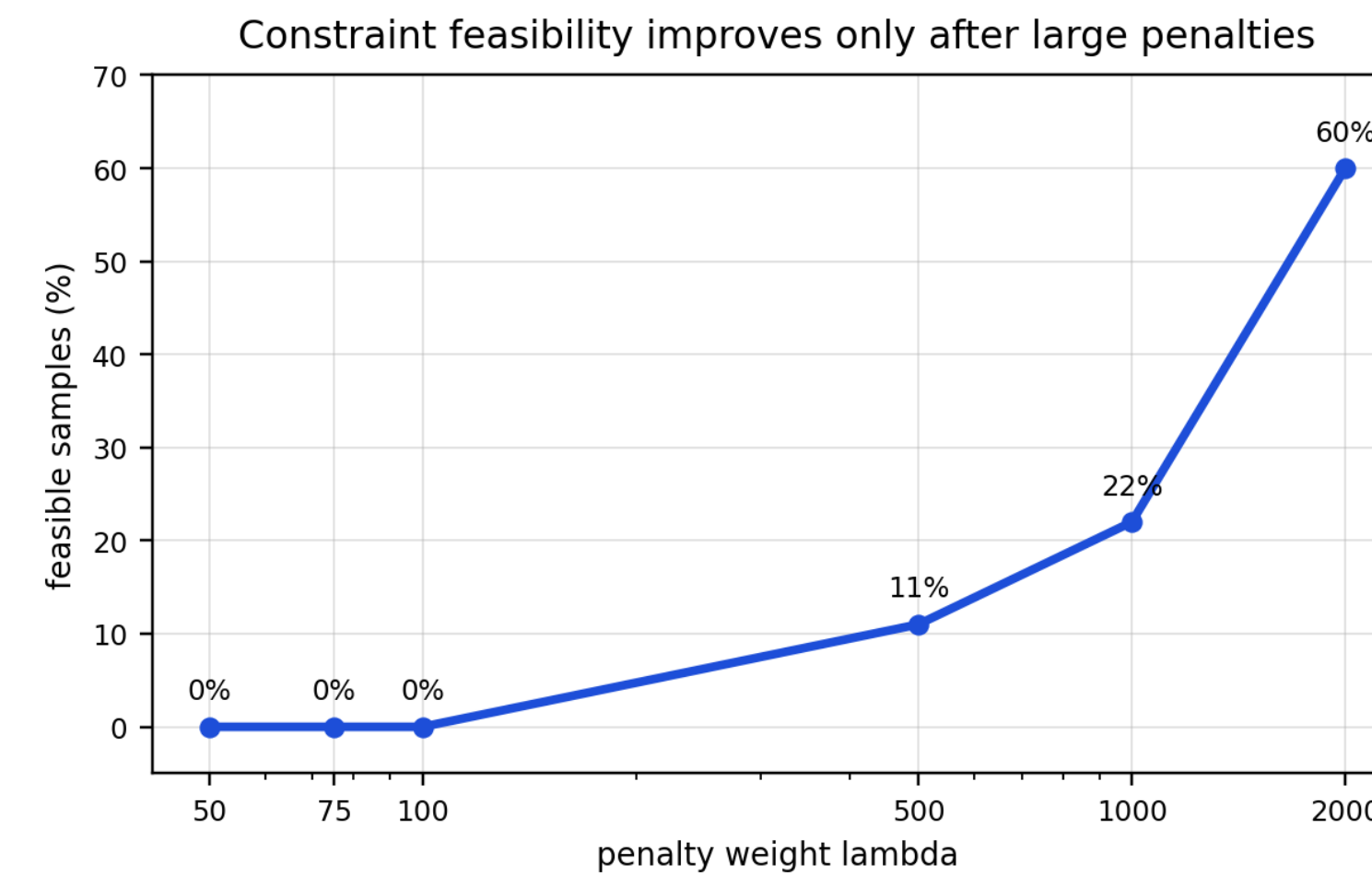
9 Results: 8 x 8 instance

All methods are evaluated on the same problem instance. Feasibility is verified after decoding, not inferred from solver energy.

Metric	A*	SA	Neal
Soft cost	317	303	496
Feasible samples	100%	100%	22%
Reaches target	yes	14%	8%
Visibility	80%	-	-
Runtime / sample	10 ms	47.8 s	0.26 s

Headline: A* and SA produce comparable feasible quality, but A* is about 4,700 times faster. Neal is fast but frequently infeasible.

10 Constraint-exploitation curve



Feasibility rises from 0% to 60% only when penalties increase from 50 to 2000. A single penalty setting can misrepresent solver performance.

11 Takeaways

Formulation value
The HUBO is useful as a quantum-compatible benchmark, not as a speed win on current hardware.

Fair comparison
Penalty sweeps and direct constraint checks prevent false quantum or annealing advantages.

Quantum sanity check
On the 8-variable instance, A*, SA, Neal, and QAOA all reach the same optimum: $E^* = -118.17$. QAOA is correct but much slower.

Core message: feasibility is a first-class metric for constrained quantum optimization.

12 Next steps and contact

- Extend from static to moving targets.
- Run larger studies on real D-Wave hardware.
- Compare against continuous MPC on shared scenarios.
- Test QAOA variants and hardware-efficient encodings.

References: Masnavi et al. on continuous multi-convex MPC; D-Wave Ocean/Neal; Qiskit QAOA.
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